

CSA0927 - PROGRAMMING IN JAVA

NAME : G.RAJU

REG NO : 192411065

FACULTY : S.PUSPHALATHA

1.

The screenshot displays a Java IDE interface. On the left, a sidebar lists seven questions (Q1 to Q7) with their respective topics and marks. The main workspace is titled 'Palindrome number/string'. It features a 'Your Code' editor with a Java program that checks if a number is a palindrome. The program uses a Scanner to take input, reverses the digits, and compares the original number with the reversed one. The 'Input' field contains the number '125', and the 'Output' field displays 'Not Palindrome'. At the bottom of the workspace, there are 'Run' and 'Save' buttons.

Q1 Palindrome number/string 10 marks

Q2 First n perfect numbers 10 marks

Q3 Decimal number equivalent to binary number and octal number 10 marks

Q4 Nth max number and nth min number in array then sum of it and difference of it 10 marks

Q5 Find the average positive and negative numbers upto -1 10 marks

Q6 Removing duplicates in array 10 marks

Q7 Matrix multiplication 10 marks

Palindrome number/string

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int original = n, rev = 0;
7         while (n != 0) {
8             int digit = n % 10;
9             rev = rev * 10 + digit;
10            n = n / 10;
11        }
12        if (original == rev)
13            System.out.println("Palindrome");
14        else
15            System.out.println("Not Palindrome");
16
17        sc.close();
18    }
19 }
```

Input

125

Output

Not Palindrome

Run Save

2.

Q2 ✓
First n perfect numbers
10 marks

Q3 ✓
Decimal number equivalent to binary number and octal number
10 marks

Q4 ✓
nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5 ✓
Find the average positive and negative numbers upto -1
10 marks

Q6 ✓
Removing duplicates in array
10 marks

Q7 ✓
Matrix multiplication
10 marks

Your Code

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int count = 0;
7         int num = 2;
8         while (count < n) {
9             int sum = 0;
10            for (int i = 1; i <= num / 2; i++) {
11                if (num % i == 0) {
12                    sum += i;
13                }
14            }
15            if (sum == num) {
16                System.out.print(num + " ");
17                count++;
18            }
19            num++;
20        }
21        sc.close();
22    }
23 }

```

Run Save Draft saved

Input

4

Output

6 28 496 8128

3.

Palindrome number/string
10 marks

Q2 ✓
First n perfect numbers
10 marks

Q3 ✓
Decimal number equivalent to binary number and octal number
10 marks

Q4 ✓
nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5 ✓
Find the average positive and negative numbers upto -1
10 marks

Q6 ✓
Removing duplicates in array
10 marks

Q7 ✓
Matrix multiplication
10 marks

Matrix multiplication

Your Code

```

1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a[][]=new int[2][2];
6         int b[][]=new int[2][2];
7         int c[][]=new int[2][2];
8         for(int i=0;i<2;i++){
9             for(int j=0;j<2;j++){
10                a[i][j]=sc.nextInt();
11            }
12        }
13        for(int i=0;i<2;i++){
14            for(int j=0;j<2;j++){
15                b[i][j]=sc.nextInt();
16            }
17        }
18        for(int i=0;i<2;i++){
19            for(int j=0;j<2;j++){
20                for(int k=0;k<2;k++){
21                    c[i][j]+=a[i][k]*b[k][j];
22                }
23            }
24        }
25        System.out.println("result matrix:");

```

Input

1 2
3 4
5 6
- -

Output

result matrix:
19 22
43 50

4.

10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4 ✓
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5 ✓
Find the average positive and negative numbers upto -1
10 marks

Q6 ✓
Removing duplicates in array
10 marks

Q7 ✓
Matrix multiplication
10 marks

6/7 answered

```

1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         System.out.println("enter the size of the array:"+n);
7         int a[]=new int[n];
8         for(int i=0;i<n;i++){
9             a[i]=sc.nextInt();
10        }
11        System.out.println("enter the elements:");
12        for(int i=0;i<n;i++){
13            System.out.print(" "+a[i]);
14        }
15        System.out.println("\narray after removing duplicates:");
16        for(int i=0;i<n;i++){
17            int count=0;
18            for(int j=0;j<i;j++){
19                if(a[i]==a[j]){
20                    count++;
21                    break;
22                }
23            }
24            if(count==0){
25                System.out.print(a[i]+" ");
26            }
27        }
28        sc.close();
29    }

```

5
1 2 2 3 5

Output

```

enter the size of the array:5
enter the elements: 1 2 2 3 5
array after removing duplicates:1 2 3 5

```

5.

Find n perfect numbers
10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4 ✓
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5 ✓
Find the average positive and negative numbers upto -1
10 marks

Q6 ✓
Removing duplicates in array
10 marks

Q7 ✓
Matrix multiplication
10 marks

6/7 answered

```

1 import java.util.Scanner;
2 public class AveragePosNeg {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         double sumPos = 0, sumNeg = 0;
6         int countPos = 0, countNeg = 0;
7         while (true) {
8             int n = sc.nextInt();
9             if (n == -1) break;
10            if (n > 0) {
11                sumPos += n;
12                countPos++;
13            } else if (n < 0) {
14                sumNeg += n;
15                countNeg++;
16            }
17        }
18        if (countPos > 0) {
19            System.out.printf("Average of positive numbers: %.2f\n", sumPos / countPos);
20        } else {
21            System.out.println("Average of positive numbers: 0");
22        }
23        if (countNeg > 0) {
24            System.out.printf("Average of negative numbers: %.2f\n", sumNeg / countNeg);
25        } else {
26            System.out.println("Average of negative numbers: 0");
27        }
28        sc.close();
29    }

```

12
5
-3
4

Output

```

Average of positive numbers: 7.75
Average of negative numbers: -5.50

```

6.

QUESTIONS

Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to
binary number and octal
number
10 marks

Q4
nth max number and rth min
number in array then sum of it
and difference of it
10 marks

Q5
Find the average positive and
negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q3 Decimal number equivalent to binary number and octal number10 marks

Decimal number equivalent to binary number and octal number

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter a decimal number: ");
6         int decimal = sc.nextInt();
7         String binary = Integer.toBinaryString(decimal);
8         String octal = Integer.toOctalString(decimal);
9         System.out.println("Decimal number: " + decimal);
10        System.out.println("Binary equivalent: " + binary);
11        System.out.println("Octal equivalent: " + octal);
12        sc.close();
13    }
14 }
```

Input

10

Output

```
Enter a decimal number: Decimal
number: 10
Binary equivalent: 1010
Octal equivalent: 12
```

Run

Save

7.

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=754`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subtitle "JAVA PROGRAMMING", and a user profile for "GANGULA RAJU" with ID "1924110658D".

On the left, a progress indicator shows "3/3 answered". The main area contains a code editor with the following Java code:

```
3 public static void main(String[] args) {
4     Scanner sc = new Scanner(System.in);
5     System.out.print("Enter Income: ");
6     double income = sc.nextDouble();
7     double tax = 0;
8     if (income <= 150000) {
9         tax = 0;
10    }
11    else if (income <= 300000) {
12        tax = income * 0.10;
13    }
14    else if (income <= 500000) {
15        tax = income * 0.20;
16    }
17    else {
18        tax = income * 0.30;
19    }
20    System.out.println("Income = " + income);
21    System.out.println("Tax = " + tax);
22 }
23 }
```

Below the code editor are "Run" and "Save" buttons. To the right, an "Output" window displays the results of the program execution:

```
Enter Income: Income = 120000.0
Tax = 0.0
```

The footer of the interface includes the copyright notice "© 2026 SIMATS Online Lab. All rights reserved." and a Windows taskbar at the bottom showing the date and time as "14:57 04-06-2026".